



Department of Computer Science
Human Computer Interaction Course
COMP322

*“QOU Student Portal: Schedule, Grades and Courses Tracking
Redesign using UCD”*

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1. Team and Topic

The Al-Quds Open University (QOU) is a public distance education university in Palestine. This distance education university has been established with the decree made by the Palestinian Liberation Organization in 1991, and QOU is the first open learning institution in Palestine. Being a distance education university, the portal becomes the main pillar of communication within the university. For distance learners, the portal is the source of all information, where most of the required functions are provided.

A vital subset of these services that are frequently used by students are:

1. **Student Weekly Schedule:** a functionality that allows students to view their sessions schedule. Some of the course sessions are online, others are on campus, and some sessions are given once every two weeks. This last type can be very confusing for students, especially in the current design.
2. **Courses Services:** a service that allows users to track upcoming assignments and quizzes, get information about sessions, and reach relevant links.
3. **Exams Schedule:** a service that allows users to track their exams schedule with all relevant details.
4. **Semester Grades:** a service that allows users to track their grades in the current semester.

All the above-discussed services constitute the scope of work for us from portal services. The choice of particular services is due to their popularity among students, as was revealed during user interviews. Moreover, there were 15 usability violations identified during our heuristics evaluation for these particular services. As far as teamwork planning is concerned, we have been guided by the team-working format recommended by Dr. Hisham in the project specification material. Some slight scheduling problems were experienced, which, however, were successfully overcome due to the deadline extension.

2. Heuristic Evaluation of the QOU Portal

A. Student Weekly Schedule Feature

1. Irrelevant columns for the weekly schedule feature.

Heuristic violated: H8.

Description: There are many columns that are not relevant for the weekly schedule. The table is dense and full of irrelevant data.

Rating: 1/4.

Proposed solution: The columns that describe the course can be combined in a "course details" pop-up that appears when the user clicks the course name. Columns related to faculty members can be treated the same way.

Screenshot: [violation1.png](#)

2. Inconsistent values for the "hall" column in online course rows.

Heuristic violated: H4.

Description: Courses whose sessions are held online have inconsistent values for the hall column, sometimes it reads "online session," sometimes the cell stays empty.

Rating: 2/4

Proposed solution: There must be a unified value for the hall column for all online courses.

Screenshot: [violation2.png](#)

3. "Odd" and "Even" weeks courses.

Heuristic violated: H2, H6, H7, H8.

Description: QOU has courses whose sessions are held in specific weeks of the semester. The current design divides the semester weeks into "even" and "odd" and students must keep track of whether the week is even or odd because the sessions differ based on the week. Students have no way to know from the portal alone which type the current week is.

Rating: 4/4.

Proposed solution: The schedule must show only the current week's sessions — only the sessions that will actually be held in the current week, regardless of whether it is odd or even. The user should be able to search for the schedule of a specific future date. This way the user never needs to worry about odd/even week memorisation — the entire calculation is abstracted from the user.

Screenshot: [violation3.png](#)

4. "Discussion sessions" section.

Heuristic violated: H2, H8, H10.

Description: There is a section named "Discussion sessions" in the schedule screen. It contains no data, there is no clear meaning for what "Discussion sessions" are, and there is no documentation explaining the term.

Rating: 1/4.

Proposed solution: Either hide that section when it is empty and provide documentation on what it means, or move that section to a more appropriate location.

Screenshot: [violation4.png](#)

B. Courses Services Feature

5. Independent services section for each course.

Heuristic violated: H8.

Description: This page is supposed to provide services relevant to courses. Each course has a section of services identical to the other courses, so the screen becomes very dense. When a

student is trying to reach a service for a certain course, they must scroll and search for that course's section.

Rating: 3/4.

Proposed solution: Since the course services section is identical for all courses, there must be only one services section and the course name must become a dropdown menu to select which course to view services for.

Screenshot: [violation5.png](#)

6. The abbreviation "س.م" is unclear.

Heuristic violated: H2, H10.

Description: The meaning of "س.م" is not entirely known to some students, especially those who are relatively new to the portal.

Rating: 1/4.

Proposed solution: Documentation should be provided, or the system must use a term that is closer to the vocabulary of students.

Screenshot: [violation6.png](#)

7. Important information is missing Heuristic violated: H1, H10.

Description: The portal fails to clarify the status of course meeting information. When a student comes across an empty time slot or an empty room field, they have no way of knowing what it means. This causes significant confusion for new students.

Rating: 3/4.

Proposed solution: The system should not leave time or room fields unexplained. It should display a clear message such as: "The meeting time has not been scheduled yet" or "This is an online course and does not have a fixed face-to-face meeting."

Screenshot: [violation7.png](#)

8. The user must open exams and assignments panels for each course one by one

Heuristic violated: H7, H5, H1.

Description: It is not clear whether there are upcoming exams or assignments until the student opens the exams and assignments panel for each course one by one. This is time consuming, frustrating, and can lead to a student not knowing about an upcoming quiz or assignment.

Rating: 3/4.

Proposed solution: A three-state design for that panel: grayed out when there are no upcoming exams or assignments; green when there are upcoming items the student has already seen; a red dot indicator when there are upcoming items the student has not yet seen.

Screenshot: [violation8.png](#)

C. Exams Schedule Feature

9. The exam type options are unclear.

Heuristic violated: H2, H10.

Description: Some exam options are written in a way that may confuse students. For example, the option "Final theoretical and incomplete" combines two different exam types in one option. The student may not know whether this refers to the regular final exam, incomplete exams, or both at the same time.

Rating: 3/4.

Proposed solution: Exam type labels must be simple, self-explanatory, and require no prior knowledge of administrative terminology.

Screenshot: [violation9.png](#)

10. Exams schedule browsing and filtering issue **Heuristic violated:** H7, H8.

Description: When a student wants to see their scheduled exams, they must enter the exam type, select the semester, and press "show" before getting any results. No exams are shown by default.

Rating: 3/4.

Proposed solution: When the student opens the Exams Schedule, they should immediately get a list of all upcoming exams in the current semester. Filtering by course or exam type should be available, but the filtering process should not be a required precondition to viewing the list.

Screenshot: [violation10.png](#)

11. Irrelevant columns for the Exams Schedule feature.

Heuristic violated: H8.

Description: There are many columns that are not relevant for the exams schedule. The table is dense and full of irrelevant data.

Rating: 1/4.

Proposed solution: All irrelevant columns visible in the screenshot must be removed.

Screenshot: [violation11.png](#)

12. "Exam review" column that only works for a limited time.

Heuristic violated: H5, H7.

Description: There is a column named "Exam Review." Each exam row contains an icon for this feature, but it only works for a short time interval after the exam. If a student clicks the icon before or after that window, a new page opens with only an error message. There is no visual indication of whether the review window is currently open or closed. **Rating:** 3/4.

Proposed solution: When the revision window is open, the icon works normally. When it is closed, the icon must be visually changed or grayed out so students do not keep attempting to access exam reviews outside the available window.

Screenshots: [violation12.png](#), [violation12 Error message.png](#)

D. Semester Grades

13. The user must click "show grades" for each course one by one.

Heuristic violated: H7, H5, H1.

Description: The student cannot find out whether any grades have been published until they click "show grades" for each course one by one. This is time consuming, frustrating, and leads students to check repeatedly every few minutes during exam periods.

Rating: 3/4.

Proposed solution: A three-state design for the "show grades" button: grayed out when no grades have been published; green when grades have been published and the student has already seen them; a red dot indicator when grades have been published that the student has not yet viewed.

Screenshot: [violation13.png](#)

14. Irrelevant column for the Semester Grades.

Heuristic violated: H8.

Description: The "course ID" column is not relevant for the semester grades view since students identify courses by name, not by ID.

Rating: 1/4.

Proposed solution: The "course ID" column must be removed.

Screenshot: [violation14.png](#)

15. Predefined and static academic assessment names.

Heuristic violated: H2, H10.

Description: There is a static set of predefined academic assessment names applied across all courses. In other words, all courses display the same number and names of assessment slots regardless of the nature of the course. Students cannot tell which assessment name corresponds to which exam, assignment, or project they actually completed. **Rating:** 3/4.

Proposed solution: The assessment names and count per course must be dynamic and specific to that course. Ideally, the instructor names the assessments using labels agreed upon with students at the start of the semester.

Screenshot: [violation15.png](#)

E. Heuristic Evaluation Results Summary:

Four patterns of usability problems were discovered through the heuristic evaluation. These patterns recur in all four sections of the portal. These problems include: dense screens overloaded with unnecessary data; inappropriate use of terminology; inconsistency in representation of information throughout the interface; and a total lack of visibility on how the status has changed: no notification appears when the grade has been posted, the assignments have been posted, an exam scheduled, or an examination revision period opened or closed. The users must check for themselves constantly, hoping there will be any change.

Number three pattern the week/odd even scheduling system can be considered the most critical one. Not only is this a usability issue but also a flaw in the entire process design that is bound to produce errors due to human factors, relying solely on the user's memory, and offering no possibility for verification of which schedule applies on any given day.

3. Exploratory User Interviews

Interview Method

A total of three exploratory interviews have been performed with QOU current students by making use of the interview guide based on the model provided in Appendix B. Each interview took around 15 minutes and was either done face-to-face or over a direct discussion in the Palestinian dialect of Arabic. The participants have been informed prior to the start of the interview that this is not a test of them, there are no right or wrong answers and that they can skip any questions at their will. Prior to the interviews, all participants have signed a consent form as per Appendix C of the specifications. Two members of the team have been present at each interview.

The guide covered three areas: warm-up questions (background, tech comfort, general portal usage frequency); core experience questions (walk me through the last time you used the portal, frustrations, moments that worked, workarounds, desired features); and context questions (device, location, who else is involved when using the portal).

Participant Overview

Code	Age	Program	Year	Employment	Device
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P1	18	Business Administration	1st year	Restaurant attendant	iPhone 11 Pro Max
P2	21	Information Technology (Cybersecurity)	3rd year	Warehouse manager	Smartphone (Android)
P3	33	Master's (various disciplines)	1st year master's	Ministry of Tourism, Finance dept.	Samsung Galaxy S23

All participants are currently enrolled at QOU. All three access the portal primarily from their phones. All three balance study with employment.

Key Findings

Finding 1: Nobody gets notified of grade release. Participants P1, P2, and P3 all answered the question in exactly the same way: they continue to log in to the portal frequently, and even twice or thrice a day during exam periods, to check if grades have been added. Nobody ever gets any notification about that. Participant 1 said, "Well, they never give an exact date or time for the grade announcement. If they were able to notify us of when the grades will be announced" (P1). "During exam periods, I logged in to the portal every day, because there's always something new," participant 3 pointed out (P3, translation of Arabic version). P2 would wait for one week after exams and begin his/her checks: "One week to ten days or so is enough time, because n then they will have to announce the grades so that we can log in and check" (P2). If asked about push notifications of grade release, P1 wouldn't think twice: "Of course, that would help us a lot" (P1).

Finding 2: The odd/even week system is highly confusing and has already caused some cases of non-attendance. Participant P2 was unequivocal when he said "Until now, I have not understood it and nobody has understood it" (P2). P2 saw the odd/even week system as a system which will cause a total mess if someone misses a class in either an odd or even week. P1 conceded that there is confusion but said it is because students think too casually about it, and used WhatsApp groups as a way of dealing with the issue. However, participant P3 does not check her time table since her classes never change; however, this also implies that she has nothing to verify the odd/even week system for herself.

Finding 3: Individual assessment grades are invisible. Only the final total appears. P2 described being unable to see his grades for individual assignments, lab sessions, or activities throughout the semester: *"We don't know our coursework grades, sometimes it doesn't show up*

on the portal" (P2). He has physically visited the registration office to ask for coursework grade information: *"We go back to the registration department and ask them: what is my coursework grade?"* (P2). At semester end, he receives only a final overall grade per course. P1 noted that his professor does post activity grades, but this is not a consistent system-level behaviour across instructors. This finding validates heuristic violation #15 (static assessment names) and #13 (grade visibility).

Finding 4: WhatsApp groups function as an informal portal substitute. All three participants rely on WhatsApp student groups to fill in information the portal does not provide. P1 described them as: *"Those groups are amazing, available 24 hours, any question answered immediately"* (P1). He uses them to confirm whether exams are online or in-person, whether grades have been posted, and how to handle portal issues. P2 uses his class group to find out when grades or assignments are posted. P3 occasionally submits exam work via WhatsApp as a backup because she fears the portal will crash and she has no submission confirmation (P3, translated from Arabic). The WhatsApp group functions as the notification system the portal does not have.

Conclusion, Finding 5: Administrative procedures involve campus visits. Participant 3 mentioned never using the petitioning and communicating functions on the portal and that she preferred physically visiting the campus in case any formalities arose (P3, translated from Arabic). In her own words: "The petitioning and correspondence must become faster because we don't have time to be being masters" (P3, translated from Arabic). Participant 2 also highlighted personally going to the registration office to get the bare minimum data about his grades.

Interview Limitations

Interviews averaged 18 minutes each, shorter than the 20–45 minute target in the guide. This limited the depth of follow-up questions on some topics. All three participants were from the same geographic area (West Bank). The sample does not include Gaza-based students, who may have different connectivity and access contexts. Full transcripts are included in the appendix.

4. Personas

Interview Synthesis and Clustering

Three interviews were conducted. The coded transcripts (30–60 noun-phrase codes each, grouped by theme) produced the following behavioural picture across participants:

Behavioural Dimension	P1 — Asir (18, Business, 1st yr)	P2 — Mohammad (21, IT, 3rd yr)	P3 — Balqis (33, Master's, Ministry employee)
Portal frustration level	Low — portal "very easy", complaints mild	High — lost money and credit hours; system-caused harm	Medium — "normal, easy, nothing special"
Usage frequency (normal period)	Medium — 2–3×/week	Low — only when triggered by exam or lecture	Medium — every 2 days
Usage frequency (exam period)	High — 2–3×/day	Medium — weekly grade check	Medium — daily
Tech literacy	High — trading apps, ChatGPT, iPhone power user	Very high — cybersecurity track, codes regularly	High — uses ministry e-platform, PaISAR, revenue software professionally
Real harm from system	None	Severe — paid for same course twice; credit hours lost	None
Mobile dependency	High — iPhone only	High — phone only	Medium — phone mostly, laptop at home
Session duration / intent	Short, reactive	Short, reactive	Very short (~3 min), task-specific

Cluster A (P1+P3): Similarities include positive-to-neutral portal perception, no severe system-related injuries, and focused-tasking and speedy-exit utilization of the portal (viewing grades and communications before exiting). Neither utilizes the portal for anything else other than what is intended to be done, and WhatsApp and physical visits take care of the shortcomings of the portal.

Cluster B (P2): Mohammad is unique in his experience with the portal because he faces severe frustrations, actual academic and financial damages, detailed understanding of the problems

with the portal, and a list of workarounds, which includes making a physical visit to the registration desk, keeping track of odd and even weeks, and viewing faculty emails from a different menu.

Cluster selection: Cluster B is the primary because the redesign exists to address unmet needs. Mohammad's needs are the most specific, the most severe, and the most directly tied to the 15 heuristic violations found in the evaluation. Designing for him addresses Cluster A users as well, their needs are a subset of his. Cluster A forms the secondary persona.

Primary Persona — Mohammad Ahmad

Persona role: Primary

Field	Details
Name + photo	Mohammad Ahmad . Photo: 
Demographics	21 years old. Third-year Information Technology student, Cybersecurity track, AIQuds Open University, Deir Sudan branch. Works full-time as a warehouse manager. Commutes to campus a few times a week.
Tech literacy	Very high. Writes code daily as part of his coursework. Uses programming environments and IT tools regularly. Has three years of portal experience and knows its structure in detail, including where it fails and what data it gets wrong. Accesses everything from his phone. Has never used the portal on a laptop or desktop.

Field	Details
Context of use	Always on his smartphone, at home or on campus between sessions. Every portal visit is triggered by something specific, an upcoming exam, a post-exam grade check, or a peer messaging the WhatsApp group that something was posted. He opens the portal, does one thing, and leaves. He does not browse. When the portal fails him, his next step is the registration office in person, a 40minute commute each way.
Goals (in this system)	(1) I need to know whether any grade, assignment, lab, activity, midterm has been posted for any of my courses, without having to click into each one to find out. (2) I want to look at my schedule and know exactly which classes I have this week, without having to figure out whether it is an odd or even week. (3) I need to read messages from my professors without having to go into a separate submenu to find them.

Frustrations	The odd/even week scheduling system is the dominant pain point. He says directly: <i>"Up until now I haven't understood it, and nobody has understood it"</i> — and has attended classes on the wrong week as a result. Individual assessment grades (assignments, lab work, activities) do not appear in the portal; only the final total shows at the end of semester. He has gone to the registration office in person to find out what his coursework grade was. Faculty messages are buried: he must enter the messages section and then specifically choose the faculty tab to see what professors sent — messages that sometimes contain assignment details. He has paid for the same course twice due to a portal registration error, and was among the students who lost credit hours when the system was migrated two years ago. He has low trust in the portal's accuracy and doublechecks important information manually.
Representative quote	<i>"Up until now I haven't understood [the odd/even system], and nobody has understood it."</i> (P2)
Evidence trail	Primary source: P2 (Mohammad Amin). Corroborated by P1 on mobile-first access, WhatsApp group reliance, and reactive portal usage. Themes embodied: odd/even week confusion (violation #3, rated 4/4), grade

Field	Details
	invisibility (violations #8 and #13), faculty message burial (violation #7), no proactive notification (H1), and system-caused real harm.

Design Implications:

1. The Semester Grades page must show grade publication status for all enrolled courses on a single screen — published, not yet published, published but not yet viewed — without requiring the user to expand each course panel individually. Mohammad must be able to open grades and know in seconds whether anything is new.
2. The Weekly Schedule must compute which sessions fall in the current calendar week and show only those. The odd/even label must never appear to the user. The system must do the week-type calculation internally, not outsource it to student memory.
3. Individual assessment grades — assignments, activities, labs, midterms — must all appear within a single course view, not only at the end of semester as a total. Mohammad must not need to visit the registration office to find out how he is doing during the semester.
4. Faculty messages must be surfaced in a single notification space visible from the main portal view — not in a submenu that requires two levels of navigation to reach.
5. A grade field showing "0" before final grades are submitted must be explicitly labelled "Not yet published" or equivalent. The system must not allow a real zero and an unpublished grade to look identical.
6. The portal must never require a physical office visit to retrieve information that already exists in the system — grade breakdowns, coursework components, section availability.
7. On any screen where the schedule is relevant, the portal must communicate today's date and contextualise it clearly. The student must not need to count weeks from the semester start date.

Secondary Persona — Sujood

Persona role: Secondary

Field	Details
Name + photo	Sujood(fictional name). Photo: 
Demographics	33 years old. Master's student at Al-Quds Open University. Works full-time at the Palestinian Ministry of Tourism, Finance and Revenue department. Married with two children. Based in the Ramallah area.
Tech literacy	High. Uses multiple institutional digital systems at work every day — a ministry elicensing platform, PalSAR/Beesan software, and a specialised revenue management program. Comfortable with formal digital workflows. Uses her Samsung Galaxy S23 for most tasks; switches to her laptop at home when the task requires it. Does not explore unfamiliar portal features — goes directly to what she needs.
Context of use	Portal sessions happen in short windows — during a five-minute work break or after her children are in bed at night. She spends about three minutes per session: checks grades, scans messages, and exits. She visits campus in person roughly once a week, frequently to handle administrative matters she cannot complete reliably through the portal. During exam periods, she sometimes sends her submitted work via WhatsApp as a backup because the portal provides no confirmation that her submission went through.

Field	Details
Goals (in this system)	(1) I need to see my grades, any unread messages, and anything pending — all in one place, in under three minutes, without clicking
	through multiple sections. (2) I want to submit petitions and correspondence online without having to go to campus in person. (3) I need to know, with certainty, that any document I submit through the portal has actually been received.
Frustrations	Administrative processes through the portal are too slow and unreliable for a working mother with a master's workload: <i>"The petitions and correspondence need to be faster — since we master's students don't have time"</i> (P3, translated from Arabic). She ends up going to the university in person instead, which costs time she does not have. The portal has not meaningfully changed in over two years: <i>"Two and a half years, nothing changes"</i> (P3, translated from Arabic). She submits exam work via WhatsApp as a backup because she fears the portal will freeze and her submission will be lost — not because it always fails, but because the portal gives no confirmation receipt.
Representative quote	<i>"Three minutes — I check quickly and leave."</i> (P3, translated from Arabic)

Field	Details
Evidence trail	Primary source: P3 (Balqis Barghouthi). Corroborated by P1 (Asir) on mobile-first grade-checking behaviour, WhatsApp group reliance as informal backup, and reactive portal access. Themes embodied: short taskfocused sessions with no room for navigation errors, administrative process gaps (violations #7, #8, #10), slow portal update cycle, submission reliability anxiety.

Design Implications:

1. The landing screen of the portal must ensure that grade status, unread messages, and pending actions are made available right away without any need for further navigation. Lina is only given three minutes to complete the task..
2. Any document or exam submission must generate a visible receipt — file name, submission time, course, and a confirmation indicator. Users should not feel the need to send a WhatsApp backup because the portal gave no indication that anything was recorded.
3. Department and faculty announcements must appear in a prominent location on the main portal view or the grades view — not buried in a messages submenu that requires a separate navigation step.
4. The portal must load primary content (grade table, message count) within an acceptable time on a mobile data connection. A user on a short work break cannot wait for a slow page render.
5. Navigation paths must remain consistent across portal updates. Lina uses the portal by muscle memory — a layout change during an update forces her to spend her three minutes reorienting instead of getting information.

5. Goal Scenarios

S1 — Zeros With No Meaning

Sunday evening. It is the tenth day since Mohammad took his midterm exam in Networks Security, one of his courses at the Deir Sudan branch of his college. He logs into the portal through his mobile phone while having dinner in his work warehouse. He looks up his grade records in Networks Security. All the grades recorded under his assessment are zeros. He does not know whether the zeros represent lack of recording by the professor or whether he got zero marks. He closes the box and tries again – still no change. He joins the class WhatsApp group and writes: "Is anybody's Networks grade out?" Two classmates reply that their grades are 0 too. Another says she checked at the registrar's office, and she was told that grades had not been released yet.

Claims:

- The system must visually distinguish "grade not yet published" from "published grade of zero" on every assessment entry — a numeric 0 and an absence of grade cannot look identical.
- The system must display grade publication status for all enrolled courses on a single screen, without requiring the user to expand individual course panels to find out whether anything is new.
- The portal must not create a situation where the only reliable way to determine grade status is peer messaging or a physical office visit.

S2 — The Week That Cannot Be Named

It is Wednesday morning. Mohammad has a class in Systems Programming and another in Database. Both classes take place on Wednesdays – however, one is an "odd week only" class. Mohammad looks into the portal on his smartphone in order to find out which of his sessions takes place today. Indeed, both sessions take place every Wednesday, with week-type specification – but there is no indication of whether today is an odd or even week. In order to figure out the answer, he tries counting backwards from the first day of the semester in order to find out which week it is. He is not sure, so he sends a message to one of his classmates. Mohammad does not get a response until after he has already boarded the bus. After arriving at Deir Sudan campus, he sees that the laboratory room is empty. He had come at the wrong week. His classmate told him about it in the WhatsApp group later. **Claims:**

- The schedule view must show only the sessions scheduled to meet on the current calendar date — computed automatically from the academic calendar — so the user never needs to calculate odd/even week status themselves.
- The odd/even label must be abstracted away from the user entirely. The system must answer "what do I have today?" not "what is the pattern, and which pattern applies this week?"
- After a session passes, the portal must provide enough contextual information for a student to understand what occurred that day and what, if anything, they missed.

S3 — Three Minutes, No Answers

It is now Tuesday afternoon. Lina is sitting at her desk at the Ministry. It's five minutes before her meeting starts. She accesses QOU through her Samsung phone. She sees that there are no values in her Assessment entries to interpret. She doesn't know whether the quiz grade has not yet been uploaded or whether she has accessed the wrong page. Checking for messages, she sees that none have come through in a while, the most recent one dating back two months ago. She ponders whether she missed out on some announcement by her department that was made elsewhere in the portal. Her meeting alert pops up. She closes the application without having found what she sought for, intending to try again using her laptop later at home when her kids would be asleep.

Claims:

- Grade entry status must be immediately clear on page load — the user must not be left uncertain whether empty rows mean "not yet published," "wrong semester selected," or "no assessments scheduled."

- Unread messages and recent department announcements must be surfaced prominently from the portal's main or grade view — not in a separate messages section the user must navigate to as a distinct step.
- The portal must surface all recent announcements from a single visible location so that a user with limited time does not need to check multiple sections to be confident nothing was missed.

6. Usability Requirements

#	Requirement	ISO Goal	Source
R1	A registered student shall view the publication status of all grades for all enrolled courses in a single semester view — without expanding individual course panels — completing the check with 0 navigation errors.	Effectiveness (E)	S1
R2	A registered student shall determine which of their enrolled courses meets today using the schedule view in under 45 seconds, without manually calculating whether the current week is odd or even.	Efficiency (F)	S2

R3	A registered student shall successfully identify all unread messages and pending grade publications during a single portal session of under 3 minutes,	Efficiency (F)	S3
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#	Requirement	ISO Goal	Source
	with at most 1 wrong navigation step.		
R4	A first-time user of the Semester Grades section shall correctly distinguish a "not yet published" grade entry from a "published grade of zero" without assistance or additional navigation, in at least 9 out of 10 attempts.	Learnability (L)	S1

R5	When an assessment grade has not yet been submitted by the instructor, the system shall display the label "Not yet published" (or equivalent) instead of "0" or an empty cell in 100% of cases — a numeric zero shall appear only when a real grade of zero has been entered.	Error Tolerance (R)	S1, S2
R6	A registered student shall locate and read the most recent unread faculty or department message without entering a dedicated messages submenu — reaching it directly from the main grade or	Effectiveness (E)	S3
#	Requirement	ISO Goal	Source
	schedule view — in under 30 seconds.		
R7	The redesigned QOU portal shall achieve a SUS score of 70 or above, averaged across at least 3 think-aloud test participants, in the Phase 3 evaluation.	Satisfaction (S)	S1, S2, S3

Appendix — Interview Transcripts

The full transcripts of the interviews conducted with all three individuals (P1, P2, P3) can be found in the appendices submitted with this report. These include the complete translations, code lists of noun phrases categorized according to themes, descriptions of important themes, and analyses of behavioral variables used in developing the personas mentioned above. Anonymized versions of the transcripts have been provided, where the names of the participants have been substituted by the codes (P1, P2, P3).

Appendix C: Participant consent form

A consent form must be signed by every interview and usability test participant before the session begins. This template may be used directly or adapted. Teams are encouraged to translate it into Arabic for Arabic-speaking participants; both versions can be offered.

COMP322: Human Computer Interaction

Participant Consent Form

Project title: QOU Student Portal: Schedule, Grades and Courses Tracking Redesign using UCD

Team members: Osama Manasrah ,Sara Ahmad ,Eman Ladadwah , Donia Assi

Course instructor: Dr Hisham Ihshaish, Department of Computer Science, Birzeit University

What is this study about?

We are students in a Human-Computer Interaction course at Birzeit University. As part of our course project, we are studying how people use [insert system or service name] and looking for ways to improve it.

What will you be asked to do?

You will be asked to (choose one):

- Take part in an interview of approximately 20-45 minutes about your experience with the system.
- Use a prototype of a redesigned system while we observe, for approximately 30 minutes, and answer a short questionnaire afterwards.

We may take written notes, and with your permission, record audio or video. Recordings will only be used by our team for analysis and will be deleted at the end of the course.

What are the risks and benefits?

There are no known risks beyond those of a normal conversation or software use. There is no direct benefit to you, but your input will help us design a better version of the system.

How will your information be used?

- Your identity will not appear in any report or presentation. We will refer to you by a code such as "Participant 1".
- Direct quotes may be used, but only in ways that cannot identify you.
- Data will be stored securely and accessible only to the team and the instructor.
- All raw data will be deleted at the end of the academic semester.

Your rights

- Participation is entirely voluntary.
- You may decline to answer any question.

- You may withdraw at any time without giving a reason and without any consequence. Any data already collected from you will be deleted.

Consent

I have read and understood the information above. I agree to take part in this study.

- ☒ I agree to participate.
- ☒ I agree to be audio or video recorded.
- ☒ I agree to have direct quotes used anonymously in the team's report and presentation.

Participant name: Mohammad Ahmad

Signature: 

Date: 22/5/2026

Team member witness: Enan, Dania

already collected from you will be deleted.

Consent

I have read and understood the information above. I agree to take part in this study.

- ☒ I agree to participate.
- ☒ I agree to be audio or video recorded.
- ☒ I agree to have direct quotes used anonymously in the team's report and presentation.

Participant name: Sujood

Signature: 

Date: 22/5/2026

Team member witness: Sara, Osama

- احنا مجموعة طلاب من جامعة بيرزيت بنعمل تحليل لبوابة جامعة القدس المفتوحة واحنا حابين نتعرف عليك
- = بلقيس برغوثي خريجة جامعة بيرزيت وبتدريس ماستر في جامعة القدس المفتوحة وبشتغل بالسياحة في وزارة السياحة (قسم المالية والارادات)
- كم عمرك
- = 33
- = عندي بنتين
- شو علاقتك بالتكنولوجيا؟ شوب تستخدم من تطبيقات التكنولوجيا
- = وسائل التواصل فيس انستا وبالشغل بستخدم المنصة الالكترونية للتراخيص ويبسان وبرنامج خاص بالارادات
- نوع تلفونك؟
- = سامسونج س 23
- = كيف بتستخدم موقع الجامعة من اي جهاز؟
- من التلفون واللابتوب بس الاغلب تلفون
- كل كم من يوم بتفتحي التطبيق؟
- = كل يومين بس بالامتحانات كل يوم
- كيف تجربتك مع البوابة الجامعة؟
- = بوابة عادية سهلة ما فيها اشي مميز وما بتستخدم بشكل مستمر و في اشياء فيها ما بتفيد مثل الطلاب والمراسلات مئا في تعامل معها بشكل رسمي (بلجا للجامعة اكثر)
- اول ما استخدمتي البرنامج كان سهل؟
- = اه بس التحديث عنده بطيء في تأخير في التحديث الي سنكتين ونص نفس اشي ما بتتغير
- شو اشي ما بتحببه بالتطبيق؟
- = لا الامتحان مثلا بنزل WORD ومرات بنبعث عالواتس خوف من البوابة بلاش تعلق بس مش كثير بيعتوه من باب التاكيد
- كم مرة بتداومي بالجامعة FACE TO FACE?
- = قليل ممكن مرة بالاسبوع
- بالعادة لما تصير مشكلة بالبوابة مع مين بتواصل؟
- = مع الدعم الفني استجابة سريعة
- ليشب تفوتي عالبوابة؟
- = تشيك ايميلات وعلامات واذا في رسائل
- الجدول مش كثير بتشوفيه؟
- = لا لانه ثابت والزوجي والفردى غالبا ثابتات
- لما بتفوتي عالبوابة كم بتتعدى وقت؟
- = 3 دقائق بشيك بسرعة وبطلع
- شو بتتمنى اشي كان بالمنصة
- = الطلبات والمراسلات تكون اسرع باعتبار انه احنا ماستر ما معنا وقت
- تمام يا بلقيس يعطيك العافية

حكينا لنش بدنا بالطفلة -
بنا تعرف. طوله كم حيرك شو بتبرس =
معد امن امن امن سوان حمري 21 سنة بنرس الجامعة القدس المفتوحة تخصص تكنولوجيا المعلومات
فرع امن امن امن الى 3 سنن بالجامعة بشغل و بنرس بنفس الوقت
= شو بتشغل
-
ماتك المخلون لي شركة لشراري
= شو بتسخدم بتسخدم برامج ثالثة ؟ غير هيس بورك
-
مراجع الي بكتب عليهم اكوار
= احكلي و انت سنة اولي لما بلشت تسخدم نظام الجامعة كان عنك حد بتسافر اى احكلي انا صر لي
-
فصص غريبة بالجامعة مشغلة بالنظام
ا، كان فيه وه اول ما دخلت الجامعة كان ابن حم كان في الجامعة، نصصني وكا شي، بزل المواد وكا شي،
-
بن في اخاف بتصير بالجامعة
مثلا انا بدى اتزل مواد على الخطط الدراسية، لازم يعطوني الخطط الدراسية. كنت اتزل على الخطط
الدراسية، وعدي صر اتزل مواد، اتزل المادة برمتين وتروح على المادة، يعني القديمة
وكنت مخصصة من زمان، بن هي ماثلة عندها الخطه
فيها هو اللي بفرش الامور.
-
لما تفر النظام
الجديد، تفر فان ستن، النظام الجديد غريظ كل الامور، رواج على ساعت، مش بن على انا، وعلى الطالب
وبدل على الوالة الا لما يقى على امتحان او على محاضرة، بدخل على الوالة بن لما يكون على. كان
الامتحانات
-
حين بتزل لما يكون عليه محاضرة؟
بن يعني امتحلت او على محاضرة برجمة او شي في الكمبيوتر
-
صحي كان كم غريظا؟
لما تكون ثلث ايام او اربع ايام، لما اتوف ان على محاضرات، لما اتوف الجامعة والامتحانات مثلا
= ايما بتعرف ايما تزل الامتحانات
لما مثلا ما هكم امتحان، بعد اسبوع بدخل بشوف الامتحانات، بشوف الجامعة بعد اسبوع او اسبوع ونص بدهم
يتزلوا الامتحانات، بن تدخل بشوف الامتحانات على الوالة
خا بقدر تفح الوالة وبشوف انا الامتحانات ثالثة او الامتحانات ثالثة او الامتحانات ثالثة
=اول ما بدخلت الجامعة، ها نظام زوجي وفردى، كيف تعاملت معه؟
بن مثلا انا يكون طينا محاضرة يوم الاربعة، ثالثة في زوجي وفي. ما لحد الان ما فهمته، وما حدا فاهمه
فردى. انا ما لفتش الزوجي زوج على الفردى، واذا ما لفتش الفردى زوج على الزوجي، يعني بفرش
الامور
-
طبه، ايتو بتواصلوا كثر مع الناس على حروب الجامعة؟
-
عنا حروب نسال في الاسئلة، مثا: ايتو تزل الامتحانات، وايضا تزل الامتحانات، يعني الامتحانات ايتو عليها
وكمان عنا. الامتحانات، لازم نحل الشغل والمواد التي ايتو بدخلت الامتحانات، وفي مواد ما بدخلت
الرسالة، الجامعة بتبع رسالة، بن احيانا بتشوف انا عضو هيئة التدريس بحث رسالة او لا. يعني ايتو بغير
بشوف

=المواد المقررة، الامتحانات، ايتو ما بتسخدمها هههه يعني لما يكون في ايتي، ما بدخلت
الامتحانات؟

-
لا، ما بتسخدم الامتحانات، يعطونا ايتو رسالة نصية على الوالة
-
يعني اعمل بحث او Assignments ايتوكم=أو، وبالنسبة للامتحانات، هاي بتزوها على جدول الامتحانات
الاول، بن الثاني ما بدخلت، زى هيك Assignment ايتو شغل، برعنه بتزوها. يعني مثلا ممكن بتزوا
صبح؟
-
عدم صحح. وعاملت الاحكام ما بدخلت، مخرج قديم الامتحان بتسخدم: شو علامة الاعمال عندي؟ يعني
مراجعات ما بتزل على الوالة
بن ايتو بالهولة بدخلت علامة اليد وعلامة النهائي. اما الامتحانات، قش جنا فيها، او الامتحانات، او النهائي،
-
هاي ما بدخلت، بالهولة بتزولنا الامتحانات كاملة للامدة
= من ومن بتسخدم
-
عن الحول
= ايتي مزج عنده شو ؟
-
حالا اسبوع الزوجي واسبوع الفردي، انا ما مشيت على الخطه الدراسية صبح، ممكن تخسر ساعت، يعني لما
تجي تزل الفصل الحالي، ممكن بتزولك مواد وتلك فيها، فترم الواحد يكون مراجع الخطه، حلقها،
هاي من الاشياء التي لازم الواحد يفهمها. وكمان الرسالة التي. وكنت شو طيه، حشان ما بخسر ساعت
بمتزولنا ايتو اعضاء هيئة التدريس، لازم تدخل على هذه الرسالة وعلى اعضاء هيئة التدريس حشان تشوف
مش كل الرسالة بتكون واضحة، مرات يكون فيها تشغلات، او اخبر خريجن، او. الرسالة التي بتزولنا ايتو
-
دورات، يعني لازم تدخل بشغل على رسالة اعضاء هيئة التدريس وبشوف

=اتو بتشغل الجدول حسب ثواني عندها ثواني، شو لي عنك محاضرات، وشو الوقت المتاح لك، وفي
نيل انا ما عندهم وقت فراغ، بمتزولنا حاليه مع تذكير معين ويعطونا من تذكير لتذكير
-
ايتو انا ما بديك الموعد، بقدر تطلع او تحذف المادة
= لا تا بمتزولنا لازم ع روج شية اقلية
-
= في مشكلة صارت بديت بالنظام ؟
لما طغوا الخطط الدراسية والنظام الدراسي، كثر طاب راحت. طوبه ساعت. ولما شاعر المشاكل التي
وكمان جابوا مرشحين، يعني صر. صارت والطب صابوا مراجعهم فيها، ركعوا موضوع الـ 9 ساعت
عنا برمتين اكليمي تروح تسلمه ونصفر منه، وهو بمتزولنا على الاسئلة. يعني الحمد لله ركعوا الـ 9 ساعت

نحن مجموعة طلاب من جامعة بيرزيت ضمن مساق
ماخذين بوابه جامعه القدس المفتوحة بدنا نشوف هل البوابه بتبدي احتياجاتكم ولا لا
ونفهم طبيعته المستخدمين وحيثنا نعمل معك مقابلته لانه طالبه ماجستير
عرفينا عن حاله
بلفيس برغوثي
ماسجر ادارة وسياسات عامه في جامعه القدس المفتوحة
بشغل ب وزارة السياحه
حمري 33

بداوم بوزارة السياحه عندي بنتين
بالوزارة بشغل بقسم الماليه والمهن السياحيه الارادات
بدي اسالك عن علاقتك ب التكنولوجيا ايش بتستخدمي بالعاده

البكيش السوشل ميديا العاليه
بالشغل بستخدم منصه الالكترونيه تراخيص وبرامج المحاسبه بيسان وبرامج خاص بالارادات
نوع تلفون سامسونج
كل كم بتستخدمي البوابه الجامعيه واغلب الوقت من وين بتسخدميها من التلفون بفتحها لحد 3 او 4 مرات بالاسبوع بفترة الاضحاتان بشكل يومي

وين بتكوني لما بتفتحي الموقع
بكون بالبيت اغلب الوقت بكون حامل البنوتين وماسك التلفون ويدخل على البوابه
كيف تجربتلك بشكل عام مع البوابه
ايش صار معك مواقف مصيره او غريبه

بوابه واضحه وسهله بس هي ما بتستخدموها بشكل رسمي كثير في عنا مثلا طلبات ومراسلات هذول ما بفتح ومش ماخذينا بشكل جدي بطل الجا للجامعه بدل البوابه لو بدي ااجل فصل اسحب فصل ما بتوخذ البوابه الطلب

اول ما بلشتي تسخدمي البوابه
تغلبت فيه انه بس تحديثه بطيء ما بحدنوا البوابات ومش فعال بطول ل تتغير المعطومه
ايش اكثر اشي بترفرل ب هاي البوابه
لا لانه مش معد زياده مثلا الامتحانات ببعتها واتساب لانيهم بخافوا انه تعلق البوابه

كم مرة بدوامي وجاهي
احنا بسبب الظروف خلونا نداوم الالكتروني وبالعاده منداوم بس يوم
لما بصير معك مشكله بالبوابه ايش بتعمدي بتواصل مع الدعم الفني تلفون بشغولي بوابتي
لما تدخل على البوابه بالعاده بس تشيك ايميالات من الكاتره واذا في علامات نازله
الجدول مش كثير بشوفه لانه ثابت
زوجي والفردى عنا احنا الماستر بكون ثابت عشان هيكله ما بغيره كثير
لما ادخل على البوابه كم تقريبه بتفتحي ما بطول بشيك بسرعه تقريبا 3 دقائق او اقل
ايش بتجدي بتغير ب المنصه المراسلات الطلبات تكون اسهل واسرع لانه احنا الماستر مش زي الكالوريوس معنا وقت ننزلهم